A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

Jennifer Zhang John Wright

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Abstract

Large language model (LLM) calibration is nearly always reported as a single aggregate scalar, obscuring whether miscalibration is diffuse or concentrated in specific knowledge domains. We develop a statistically rigorous audit framework and apply it to MMLU [Hendrycks et al., 2021] using token-level log probabilities from six instruction-tuned models spanning two families and five parameter scales: GPT-4o-mini, Llama-3-8B-Instruct, and Qwen-2.5-{0.5B, 1.5B, 7B, 14B}-Instruct. Our framework combines a three-level mixed-effects model, isotonic and kernel-smoothed reliability curves, empirical Bayes shrinkage, and Benjamini–Hochberg FDR correction. Across the six models, between-subject calibration variance accounts for 1.7–7.9% of the total, and 55–57 of 57 subjects are flagged as significantly miscalibrated at FDR $\alpha = 0.05$. Subject-level ECE rankings are highly correlated for the five larger models (Spearman $\rho \in [0.78, 0.94]$, all pairs); the smallest model (Qwen-2.5-0.5B) shows extreme positional bias (predicting D 59% of the time) that produces a qualitatively different miscalibration pattern uncorrelated with the others. The same subjects are robustly worst-calibrated across the larger models (college-level quantitative STEM, niche factual recall, structured ethical/legal reasoning) and the same subjects are robustly best-calibrated (high-school-level Social Sciences and applied/factual subjects). Within the Qwen family across three matched-precision scale points (1.5B/7B/14B), both the between-subject ICC and the entropy coefficient grow monotonically with scale ($0.017 \to 0.049 \to 0.059$ and $-0.004 \to +0.029 \to +0.040$ respectively), a marginal pattern consistent with prior aggregate findings on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025].

1 Introduction

Large language models are increasingly deployed in high-stakes settings—healthcare, law, education—where the reliability of model-expressed confidence is as important as accuracy itself. A well-calibrated model is one whose stated confidence tracks its empirical accuracy: if a model assigns 80% probability to an answer, it should be correct 80% of the time. Miscalibration, particularly overconfidence, means that users who rely on model confidence as a decision signal will be systematically misled.

A substantial literature has studied LLM calibration. Guo et al. [2017] established the modern empirical study of neural network calibration, popularizing expected calibration error (ECE) as a standard metric and demonstrating the effectiveness of temperature scaling as a post-hoc correction. Kadavath et al. [2022] showed that for multiple-choice tasks, token-level log probabilities provide a meaningful calibration signal, and that base (pre-RLHF) models exhibit better-calibrated token probabilities than instruction-tuned counterparts. Xiong et al. [2024] provided a systematic benchmark of confidence elicitation methods, finding that all methods "struggle in challenging tasks, such as those requiring professional knowledge"—suggesting calibration quality may vary by domain, though the authors do not investigate this directly. Jiang et al. [2021] similarly found that question-level features such as answer entropy predict calibration quality, a pattern we examine formally here.

What is missing is a *statistical audit* of that heterogeneity. To our knowledge, the existing literature reports subject-level ECE without (i) stabilizing estimates against large differences in per-subject sample size, (ii) formally testing which subjects are robustly miscalibrated after correcting for multiple comparisons, (iii) decomposing how much total calibration variance is attributable to domain versus subject versus individual question, or (iv) characterizing subjects by their *pattern* of miscalibration rather than its scalar magnitude.

This framing is analogous to the gap between per-group accuracy reporting and a rigorous fairness audit [Barocas et al., 2023]: the fairness literature long ago recognized that single-number summaries are insufficient for subgroup auditing, and we import that logic into calibration evaluation. The downstream implication of the audit is structural: if miscalibration is diffuse, a single global recalibration (e.g., temperature scaling) is a reasonable fix; if it is concentrated in specific subject clusters and those clusters are stable across models, then subject-aware recalibration becomes a structurally appropriate scoping choice—a position that has been validated empirically by Luo et al. [2025], who fit subject-specific temperature parameters per MMLU subject. Our contribution is the diagnostic arm of that picture: stable audit findings establish whether the structural assumption behind a subject-aware correction is warranted in the first place.

Our central scientific questions, restated:

1. **Where** is calibration good and bad? Variance decomposition, FDR-corrected subject rankings, and a within-MMLU contrast between well-calibrated and poorly-calibrated subject clusters.

2. **How consistent** is the answer across models? Pairwise Spearman correlations of subject-level ECE across six models.

3. **What mechanisms** produce the observed heterogeneity? Question- level predictors, plus a within-Qwen scaling sweep across three matched- precision checkpoints (1.5B, 7B, 14B).

2 Related Work

Calibration of neural networks. Guo et al. [2017] showed that modern neural networks are systematically overconfident and that temperature scaling largely corrects this at the aggregate level, with Zadrozny and Elkan [2002] earlier establishing isotonic regression as an alternative nonparametric recalibration method. Kadavath et al. [2022] extended calibration evaluation to LLMs, establishing that token-level probabilities on multiple-choice tasks carry a meaningful calibration signal. Nakkiran et al. [2025] and Yaldiz et al. [2026] study how instruction tuning affects calibration, finding that RLHF-style fine-tuning tends to degrade calibration relative to base models, an effect also documented in Ouyang et al. [2022].

Domain-level heterogeneity. Xiong et al. [2024] is the closest prior work in spirit, benchmarking confidence elicitation across task types. Jiang et al. [2021] showed that question-level features predict calibration on QA tasks. Luo et al. [2025] fits subject-specific temperature parameters for each of MMLU's 57 subjects, implicitly acknowledging that a global correction is insufficient; Tan et al. [2026] uses base model signals for unsupervised recalibration. Neither work performs a formal statistical audit of subject-level heterogeneity.

Multiple testing and empirical Bayes. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] controls the false discovery rate and is standard in high-dimensional inference. Empirical Bayes shrinkage of group-level estimates [e.g., Efron, 2012] is standard in multilevel modeling via best linear unbiased predictors (BLUPs). We use the multilevel modeling framework of Raudenbush and Bryk [2002] to decompose calibration variance across levels of the MMLU hierarchy.

3 Data

MMLU. We use the Massive Multitask Language Understanding benchmark [Hendrycks et al., 2021], accessed via the `cais/mmlu` dataset on HuggingFace. MMLU comprises 14,042 four-option multiple-choice questions across 57 subjects nested in four broad domains: STEM, Social Sciences, Humanities, and Other. Subject sizes range from 100 to 1,534 questions (median 163). The three-level hierarchy—questions $\rightarrow$ subjects $\rightarrow$ domains—is natural for the proposed analysis. We use the test split throughout.

Models. We query six instruction-tuned models spanning two families and five parameter scales. **GPT-4o-mini** is queried via the OpenAI Batch API at temperature 0 (`max_tokens=1`, `top_logprobs=20`); the choice of 20 (rather than the API default of 5) ensures all four answer tokens are captured for virtually every question. **Llama-3-8B-Instruct** and **Qwen-2.5 {0.5B, 1.5B, 7B, 14B}-Instruct** are run locally on a Colab GPU. The 7B, 8B, and 14B models use 4-bit quantization (BitsAndBytes; Dettmers et al. 2022); the 0.5B model uses 4-bit and the 1.5B model is run twice (once at 4-bit, once at fp16) so the precision effect can be examined directly.[1] For all six models we pre-fill the assistant turn with "The answer is" before extracting logprobs at the next-token position. Test-set accuracies are 0.749 (GPT-4o-mini), 0.606 (Llama), 0.405 (Qwen-0.5B), 0.571 (Qwen-1.5B fp16), 0.685 (Qwen-7B), 0.757 (Qwen-14B). Qwen-0.5B exhibits extreme positional bias (predicting D 59% of the time vs. A 5.6%), discussed where it affects results.

4 Methods

4.1 Confidence Elicitation

For each question, we extract the token-level log probabilities over the four answer tokens (A, B, C, D). Applying softmax renormalization yields a proper probability distribution over options; the model's confidence $c_i \in (0, 1]$ is the maximum of this renormalized distribution, equal to $P(\text{predicted answer} \mid \text{answer} \in \{A, B, C, D\})$. This is the standard approach in calibration research [Kadavath et al., 2022] and avoids the noise of verbal elicitation.

For GPT-4o-mini, logprobs are drawn directly from the API's top-20 response. For the open-weight models, logprobs are extracted from the full vocabulary softmax at the assistant-generation position after the "The answer is" prefix. All pipelines compute the same quantity: $P(\hat{y} \mid \hat{y} \in \{A, B, C, D\})$.

[1] The fp16 1.5B run was added after we observed that the 4-bit 1.5B run had unusual positional bias (predictions skewed toward B at 36%) and accuracy 0.054 below the published full-precision figure. The fp16 run brought both numbers in line with the published figures and is used as the canonical 1.5B in the main analysis. The 4-bit version is preserved for the precision-effect appendix.

4.2 Outcome and Covariates

Primary outcome. The primary outcome is the *signed calibration gap* at the question level:

$$\text{gap}_i = c_i - y_i, \tag{1}$$

where $y_i \in \{0, 1\}$ indicates correctness. This quantity is positive under overconfidence and negative under underconfidence, and supports regression-based inference that binning-based ECE cannot. Its mean at the subject level directly measures the direction and magnitude of miscalibration.

Question-level covariates. We extract the following features from each question:

- **Word count**: number of words in the question stem.
- **Max option length**: word count of the longest answer choice.
- **Negation indicator**: binary, flagging presence of *not*, *except*, *least*, *never*, or *no*.
- **Entropy**: Shannon entropy in nats of the softmax distribution over A/B/C/D, $H_i = -\sum_k p_{ik} \log p_{ik}$. Zero when one option dominates, $\log 4 \approx 1.386$ when uniform.

All continuous covariates are standardized before model fitting.

Heteroskedasticity note. The variance of $\text{gap}_i = c_i - y_i$ is not constant across subjects: since $y_i \sim \text{Bernoulli}(p_j)$, its contribution to variance is $p_j(1 - p_j)$, which is highest for subjects near 50% accuracy. We check residual plots for evidence of heteroskedasticity and note this as a potential limitation of the homoskedastic mixed model.

4.3 Nonparametric Calibration Curves

For each subject, we estimate a reliability curve using isotonic regression—a nonparametric monotone fit of empirical accuracy on confidence [Zadrozny and Elkan, 2002]—rather than arbitrary equal-width binning. The monotonicity constraint is theoretically motivated but is an assumption rather than a guarantee. We therefore fit an unconstrained kernel smoother (Gaussian, $\sigma = 1.5$ bins) alongside each isotonic curve; substantive divergences between the two identify subjects where the monotonicity assumption may distort the estimated curve.

ECE per subject is computed as the weighted mean absolute deviation between the reliability curve and the diagonal, using 20 equal-width confidence bins.

4.4 Multilevel Regression and Variance Decomposition

We model the question-level calibration gap across all 14,042 questions using a three-level mixed model with questions nested in subjects nested in domains:

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \tag{2}$$

where $\mathbf{x}_{ijk}$ are standardized question-level covariates, $u_k \sim \mathcal{N}(0, \sigma^2_{\text{domain}})$ is a domain random intercept, $v_{jk} \sim \mathcal{N}(0, \sigma^2_{\text{subject}})$ is a subject-within-domain random intercept, and $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma^2_\varepsilon)$ is residual noise. The model is estimated by restricted maximum likelihood (REML) using `statsmodels.MixedLM` [Raudenbush and Bryk, 2002].

The fixed effects $\boldsymbol{\beta}$ characterize which question features are associated with miscalibration across the full dataset, interpreted as descriptive associations rather than causal effects. The variance components ($\sigma^2_{\text{domain}}, \sigma^2_{\text{subject}}, \sigma^2_{\varepsilon}$) answer RQ1 via an intraclass correlation (ICC) decomposition: the fraction of total calibration heterogeneity attributable to each level of the hierarchy. Subject-level shrunken calibration estimates are the fitted random effects $\hat{v}_{jk}$ (BLUPs), which implicitly borrow strength across subjects in proportion to sample size.

4.5 Multiple Testing Correction

For each of the 57 subjects we test H_0: mean calibration gap = 0 (no systematic miscalibration) against the two-sided alternative using a one-sample t-test on the question-level gaps. We apply the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] at $\alpha = 0.05$ to control the false discovery rate, yielding a ranked list of subjects where miscalibration is statistically established rather than just descriptively apparent.

4.6 Cross-Model Replication

The full pipeline is run on each of the six models. Subjects flagged as miscalibrated by multiple models suggest task-driven miscalibration reflecting question structure; subjects that diverge suggest model-specific effects of training. We compare subject-level ECE rankings across all 15 model pairs using Spearman rank correlation with bootstrap confidence intervals (2000 replicates, seed 42).

5 Results

5.1 Variance Decomposition (RQ1)

Table 1 reports the ICC decomposition from the three-level mixed model for the six main models. Across all six, the bulk of calibration-gap variance lies at the question level (92.1–98.3%), and domain-level variance collapses to zero. Between-subject ICCs span nearly a factor of five: GPT-4o-mini concentrates 7.9% of variance at the subject level, Qwen-14B 5.9%, Qwen-7B 4.9%, and Llama and the smaller Qwen models cluster around 1.7–2.0%.

Table 1: Variance decomposition of the calibration gap. Domain-level variance is estimated at zero for every model; $\text{ICC}_{\text{subject}}$ is the share of total variance attributable to subject-level heterogeneity.

Model	σ^2_{domain}	$\sigma^2_{\text{subject}}$	σ^2_{ε}	$\text{ICC}_{\text{subject}}$
GPT-4o-mini	0.000	0.013	0.150	0.079
Llama-3-8B-Instruct	0.000	0.004	0.190	0.020
Qwen-2.5-0.5B-Instruct	0.000	0.004	0.224	0.018
Qwen-2.5-1.5B-Instruct	0.000	0.003	0.197	0.017
Qwen-2.5-7B-Instruct	0.000	0.009	0.177	0.049
Qwen-2.5-14B-Instruct	0.000	0.010	0.156	0.059

The within-Qwen scaling pair (1.5B $\rightarrow$ 7B $\rightarrow$ 14B, all matched at the fp16/4-bit precision boundary used per model) shows ICC growing monotonically ($0.017 \rightarrow 0.049 \rightarrow 0.059$). Combined with the parallel monotonic increase in the entropy coefficient (Section 5.3), this is the cleanest within-family scaling signal in our data.

The domain-level variance is estimated at zero for every model. With only four domain groups, the likelihood is flat with respect to the domain variance component, and the estimate collapses to the boundary. We therefore report total between-subject variance without a clean domain/subject partition, and treat domain as a descriptive grouping rather than a modeled random effect.

5.2 Calibration Curves by Subject

Figure 1 shows reliability curves for the five most miscalibrated subjects by ECE for GPT-4o-mini. The curves are masked outside the observed 1st–99th percentile of confidence per subject to avoid displaying isotonic extrapolation: GPT-4o-mini's confidence is so concentrated above 0.7 that any plotted curve below this region would be pure extrapolation. Within the observed support, all five subjects show the characteristic shape of an overconfident model—empirical accuracy lying below the diagonal, sometimes sharply.

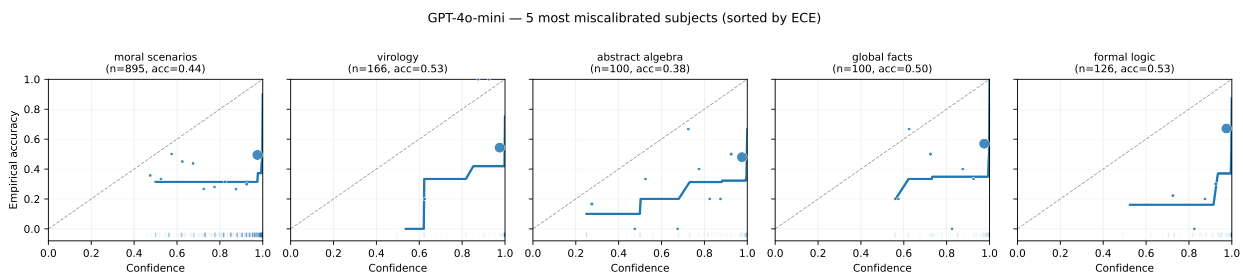

Figure 1: GPT-4o-mini reliability curves for the five most miscalibrated subjects by ECE. Solid line: isotonic regression fit, plotted only over the observed confidence range (1st–99th percentile per subject). Markers: empirical accuracy in 20 equal-width confidence bins, sized by bin count. The dashed line is the calibration diagonal. Annotations report sample size and mean accuracy.

For visual contrast, Figure 2 shows the same plot for the five *best*-calibrated subjects for GPT-4o-mini. Curves there track the diagonal closely or lie just below it; both shapes are present in the same model on the same evaluation, illustrating the heterogeneity that aggregate ECE conceals.

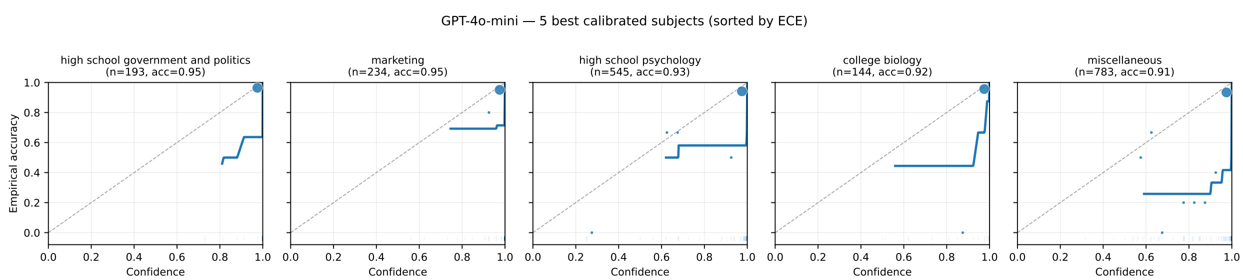

Figure 2: GPT-4o-mini reliability curves for the five *best*-calibrated subjects, shown for visual contrast with Figure 1. Same conventions.

5.3 Predictors of Miscalibration (RQ3)

Table 2 reports the fixed effects from the multilevel model across the six main models. Intercepts confirm large mean overconfidence in every model (0.179–0.233), with no monotonic relationship to scale: the Qwen intercept rises 1.5B→7B (0.179→0.233) but drops slightly at 14B (0.210). Higher accuracy at 14B (0.757 vs. 0.685 at 7B) narrows the gap even as the underlying confidence pattern grows more concentrated, and the two effects partially cancel.

Table 2: Fixed effects from the three-level mixed model for all six models. Continuous covariates standardized. Significance: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, $^{\dagger}p < 0.10$.

Predictor	GPT-4o-mini	Llama-Instruct	Q-0.5B	Q-1.5B	Q-7B	Q-14B
Intercept	0.196***	0.197***	0.209***	0.179***	0.233***	0.210***
Word count	0.031***	-0.005	-0.013*	0.003	0.019**	0.023***
Max option length	0.001	-0.001	-0.028***	0.005	-0.007	-0.007
Negation	-0.001	0.007	0.016	-0.010	-0.001	0.005
Entropy	0.016***	0.008†	-0.063***	-0.004	0.029***	0.040***

Figure 3 shows the corresponding forest plot. Word count is a positive significant predictor for GPT-4o-mini and the larger Qwens (7B, 14B), inert for Llama and the small Qwens (0.5B, 1.5B). The most informative pattern is the entropy coefficient.

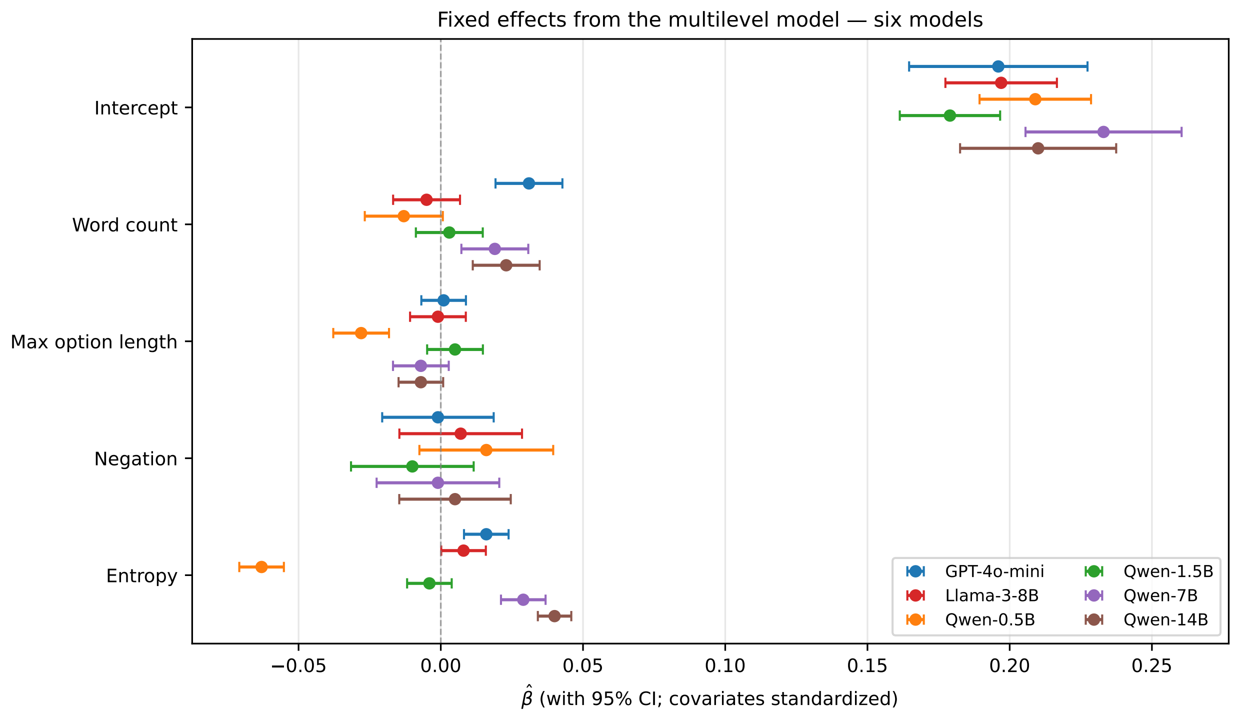

Figure 3: Fixed effect estimates with 95% confidence intervals for all six models. Estimates are on the standardized calibration gap scale. The entropy coefficient shows a striking sign progression across the Qwen scaling sweep.

Entropy: a sign progression with scale. A naïve prediction is that high option-distribution entropy reduces the calibration gap, since top-1 confidence is mechanically lower when probability mass spreads across options. Yet the entropy coefficient varies systematically across models and scales: strongly negative for Qwen-0.5B (-0.063, $p < 0.001$), near-zero for Qwen-1.5B (-0.004, n.s.), and positive and growing across larger models (Llama +0.008, GPT-4o-mini +0.016, Qwen-7B +0.029, Qwen-14B +0.040, all $p < 0.10$ or smaller). Within the Qwen family at matched precision (1.5B fp16, 7B 4-bit, 14B 4-bit), the coefficient grows monotonically: $-0.004 \rightarrow +0.029 \rightarrow +0.040$. Figure 4 traces the mechanism. Within each model we partition questions into entropy quartiles and report mean confidence (solid) and mean accuracy (dashed); the shaded area is the calibration gap.

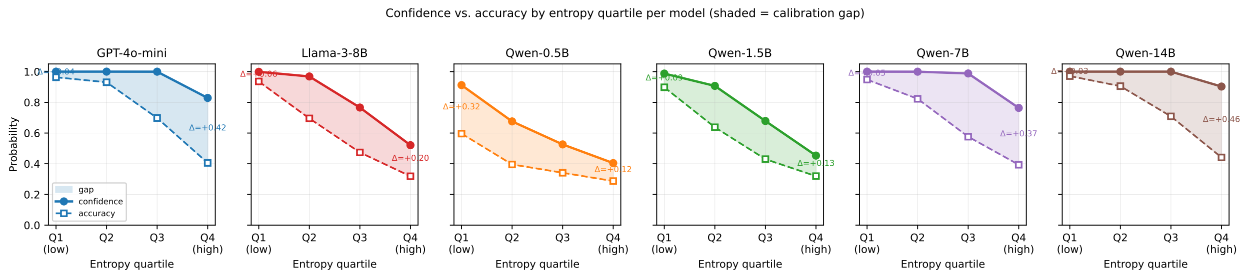

Figure 4: Confidence (solid) and accuracy (dashed) by entropy quartile per model, with the shaded gap. The pattern progresses systematically across the Qwen family: at small scale (0.5B) the gap shrinks with rising entropy; at intermediate scale (1.5B) the curves fall in parallel; at larger scales (7B, 14B) the gap widens with entropy as confidence stays near 1.0 while accuracy drops sharply.

The pattern is most striking within the Qwen family. At 1.5B confidence and accuracy fall in roughly parallel as entropy rises (gap of 0.089 at low entropy, 0.134 at high). At 7B the model holds confidence near 1.000 across the first three quartiles while accuracy drops from 0.948 to 0.576, before the model finally releases probability mass at the highest entropy quartile (confidence 0.764, accuracy 0.394, gap 0.370). At 14B this saturation extends further: confidence remains 1.000 across the first three quartiles even as accuracy falls to 0.709, and only releases marginally at the highest entropy quartile (confidence 0.903, accuracy 0.441, gap 0.462). The within-family progression is consistent with prior literature on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]: as instruction-tuned models grow more capable, they maintain top-1 confidence near saturation across an increasing range of question difficulty.

We are cautious about extending this further. The within-family scaling comparison is observational rather than a controlled mechanism test, and the Qwen-0.5B endpoint is contaminated by extreme positional bias (the model predicts D on 59% of questions). Pending replication with additional small or base models, our claim is descriptive: the four matched-precision Qwen checkpoints we examine show a sign progression in the entropy coefficient that aligns with the direction prior aggregate literature predicts.

Figure 5 summarizes the within-Qwen scaling pattern across the three matched-precision checkpoints (1.5B, 7B, 14B): both ICC and the entropy coefficient grow monotonically with scale, while the intercept peaks at 7B and dips at 14B as accuracy gains partially offset the more concentrated overconfidence pattern.

5.4 FDR-Corrected Subject Rankings

Figure 6 shows all 57 subjects sorted by mean calibration gap for each of the six main models, with FDR-rejected subjects highlighted. At $\alpha = 0.05$: GPT-4o-mini, Llama, Qwen-7B, and Qwen-14B reject all 57 subjects; Qwen-1.5B (fp16) rejects 56/57 (the lone exception is `jurisprudence`, $\bar{\text{gap}} = 0.051$, $p_{\text{adj}} = 0.18$); Qwen-0.5B rejects 55/57 (the exceptions are `high_school_statistics` and `formal_logic`).

The substantive finding is therefore the *magnitude* of heterogeneity, not the binary reject/fail-to-reject outcome. Mean gaps span a 13.0× range for GPT-4o-mini (0.037 to 0.487), 6.2× for Llama (0.069 to 0.425), and in the Qwen family 13.9× (1.5B 4-bit), 5.4× (1.5B fp16), 8.0× (7B), and 6.6× (14B). Across all six main models, the same subjects appear at the high-ECE end and the

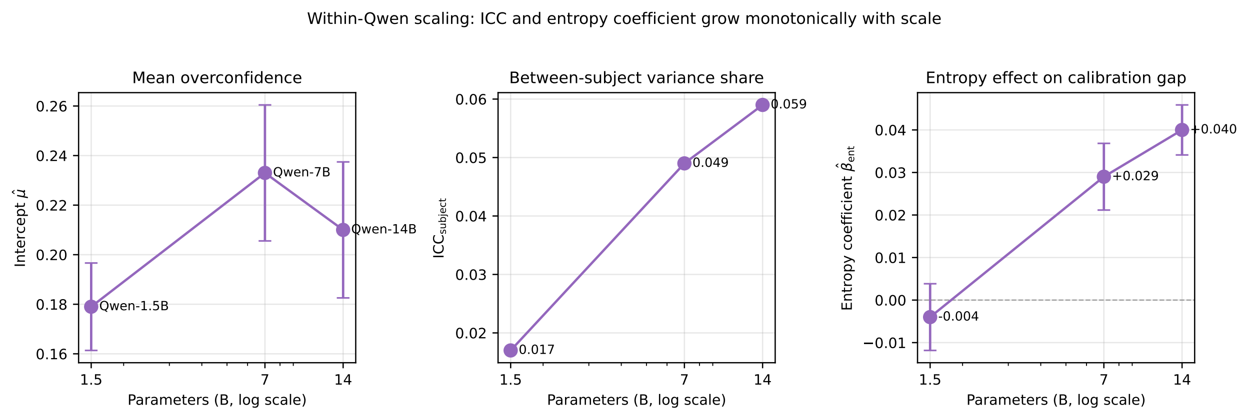

Figure 5: Within-Qwen scaling on three matched-precision checkpoints (1.5B fp16, 7B 4-bit, 14B 4-bit). Both $\mathrm{ICC}_{\mathrm{subject}}$ and the entropy coefficient grow monotonically with scale; the mean-overconfidence intercept is non-monotonic, peaking at 7B and dipping slightly at 14B as accuracy gains partially offset the underlying overconfidence pattern. Error bars are 95% CIs from the multilevel fit.

same subjects at the low-ECE end—the audit's substantive content beyond the per-model gap magnitudes.

Subjects that are robustly worst-calibrated. Across the five larger models (excluding Qwen-0.5B due to its positional-bias confound), four subjects appear in the top-10 most miscalibrated for every model: `virology`, `college_chemistry`, `college_physics`, `professional_law`. Expanding to top-15 adds `college_mathematics`, `econometrics`, `high_school_physics`, `machine_learning`, and `moral_scenarios`. These cluster into three semantic groups: (i) college-level quantitative STEM, (ii) niche factual recall (virology, global facts), and (iii) structured ethical/legal reasoning (moral scenarios, formal logic, professional law).

Subjects that are robustly best-calibrated. Symmetrically, the intersection of bottom-5 ECE across the five larger models contains `high_school_government_and_politics` and `high_school_psychology`; bottom-10 adds `high_school_us_history`, `marketing`, and `miscellaneous`; bottom-15 adds `high_school_geograp` `high_school_world_history`, `sociology`, and `us_foreign_policy`. Two patterns are visible. First, MMLU's high-school-level subjects systematically show smaller calibration gaps than their college-level counterparts; the level of specialization required appears to track miscalibration. Second, well-calibrated subjects are dominated by Social Sciences and the "Other" category (the applied/factual subjects in the MMLU taxonomy)—no STEM subject appears in the bottom-10 intersection. The audit recovers a coherent contrast: applied/factual high-school content is where calibration is reliably good, and college-level technical content is where it is reliably bad.

5.5 Sensitivity Analysis

Sensitivity-analysis conclusions extend uniformly across all six models. ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. the 20-bin baseline for every model). The Negative Log Calibration Score, a strictly proper scoring rule, ranks subjects nearly identically (Spearman ≥ 0.79 across models and bin counts). Two-level and three-level ICC estimates are numerically identical, consistent with $\sigma^2_{\mathrm{domain}}$ collapsing to zero.

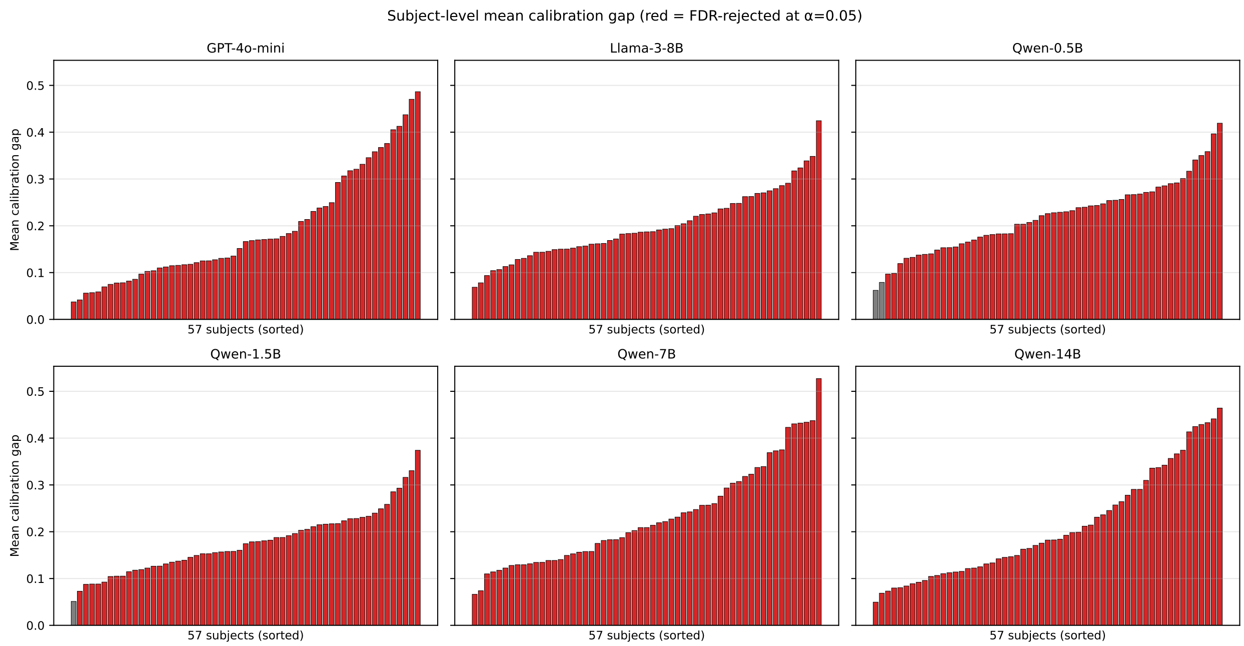

Figure 6: Mean calibration gap per subject, sorted ascending, for each of the six main models. Red bars indicate subjects where H_0: mean gap $= 0$ is rejected at FDR $\alpha = 0.05$; gray bars are not rejected.

The FDR-rejection finding is also robust to threshold choice for the four larger models: at $\alpha = 0.01$, GPT-4o-mini rejects 56/57, Llama 55/57, and Qwen-7B and Qwen-14B both reject 57/57. The smaller Qwens are more sensitive: at $\alpha = 0.01$, Qwen-1.5B (fp16) rejects 55/57 and Qwen-0.5B rejects 55/57. Qwen-1.5B (4-bit) is most sensitive, rejecting only 46/57 at the strictest threshold—a discrepancy with the fp16 version that we discuss further in the quantization-effect appendix.

5.6 Cross-Model Agreement (RQ2)

Figure 7 shows the 6×6 Spearman rank correlation matrix for subject-level ECE across the six main models. Among the five larger models, all 10 pairwise correlations fall between 0.78 and 0.94. The cross-family pair Qwen-7B vs Qwen-14B is the highest at 0.939; cross-family GPT-4o-mini vs Qwen-14B and GPT-4o-mini vs Qwen-7B are 0.92 each.

Qwen-0.5B's correlations with the other five models are dramatically lower (0.16–0.22, with bootstrap 95% CIs that include zero). This is consistent with its 10.6× positional bias (Section 3): the smallest model's miscalibration is dominated by spurious preference for the D position rather than genuine knowledge gaps, producing a distinct ranking of which subjects appear miscalibrated. We interpret this as evidence that, below approximately 1B parameters, positional bias dominates the calibration signal in our elicitation protocol; above this threshold, miscalibration patterns are shared across model families and scales.

6 Discussion

Where calibration succeeds versus fails. The two strongest findings are descriptive but coherent. Calibration is reliably good on high-school-level Social Sciences and applied/factual subjects

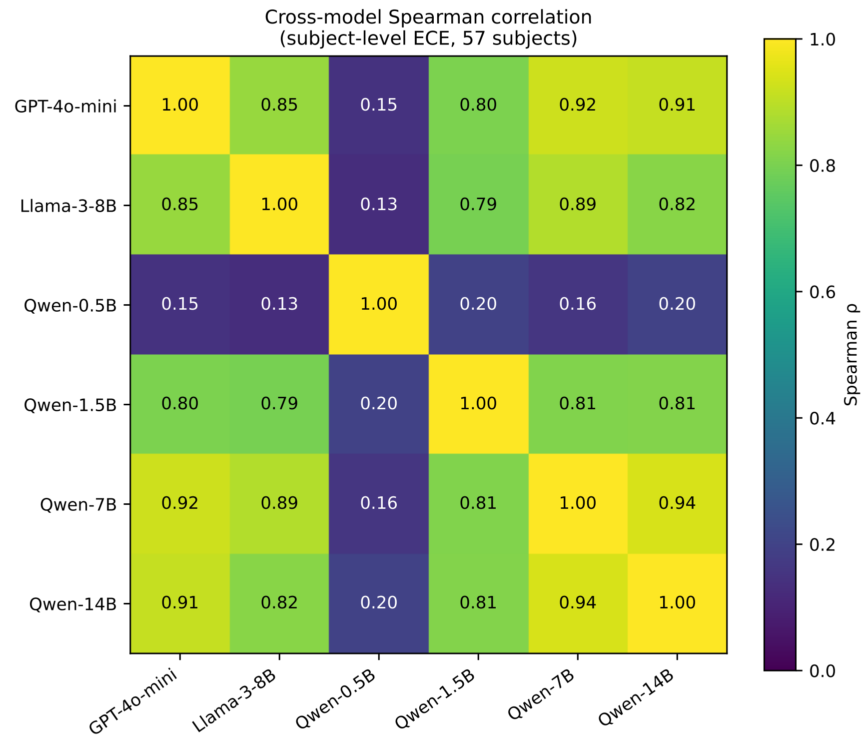

Figure 7: Spearman rank correlation of subject-level ECE across the six main models (57 common subjects). Among the five larger models, all 10 pairwise correlations fall in $[0.78, 0.94]$. Qwen-0.5B's correlations with the others are much lower (0.16–0.22), reflecting that its miscalibration pattern is qualitatively distinct due to extreme positional bias.

(`marketing`, `miscellaneous`, `high_school_government_and_politics`, `high_school_psychology`), and reliably bad on college-level technical content (`college_chemistry`, `college_physics`, `college_mathematics`, `machine_learning`), niche factual recall (`virology`, `global_facts`), and structured ethical/legal reasoning (`moral_scenarios`, `professional_law`, `formal_logic`). The same patterns appear in every model larger than 1B parameters that we examined. The level of specialization required appears to be a meaningful axis: MMLU's high-school-level subjects consistently show smaller calibration gaps than their college-level counterparts. This is the substantive finding the heterogeneity audit recovers, and the kind of structure aggregate ECE conceals.

Cross-family agreement at matched capability. Among the five larger models, the Spearman correlation between GPT-4o-mini and the larger Qwens (7B and 14B) is 0.92, exceeding even the strongest within-family pair below 14B. All 10 pairwise correlations among the five larger models fall in $[0.78, 0.94]$. Subject-level miscalibration is therefore consistent across model families and scales above 1B parameters—strong evidence that miscalibration patterns are primarily a property of the task. This stability provides the structural justification for choosing to scope corrections at the subject level rather than globally; the targets of any such correction are stable across the model space we examined, even though each correction would still be fit per-model on held-out data.

Within-family scale: a monotonic pattern in the right direction. Within the Qwen family at matched precision (1.5B fp16, 7B 4-bit, 14B 4-bit), both the between-subject ICC and the entropy coefficient grow monotonically with scale: ICC from 0.017 to 0.049 to 0.059, entropy coefficient from -0.004 (n.s.) to $+0.029$ to $+0.040$. The mean-overconfidence intercept is non-monotonic ($0.179 \rightarrow 0.233 \rightarrow 0.210$), with the dip at 14B consistent with accuracy gains partially offsetting the underlying confidence inflation. We present this as an observation about the three checkpoints rather than a controlled scaling experiment. Figure 4 traces the entropy effect to a marginal pattern in which capable models maintain near-saturated confidence even as the option distribution flattens, which is consistent with prior literature on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]. We do not claim to have identified a causal mechanism; the comparison is between checkpoints rather than a controlled within-checkpoint manipulation.

Implications for recalibration. Our framework is the diagnostic arm of a broader recalibration picture. The operational arm has been studied directly: Luo et al. [2025] fits subject-specific temperature parameters per MMLU subject and demonstrates improvements over a global correction. Our contribution is the upstream question: is subject-level scoping a structurally appropriate choice in the first place? The audit's affirmative answer—stable, cross-model targets with robust statistical support—supplies that justification. Practitioners can also reuse the audit findings as a transferable diagnostic test set: the robustly worst-calibrated subjects we identify can flag miscalibration on a new model before deployment, without re-running the full audit.

Quantization at small scale: an instructive replication. We collected Qwen-1.5B twice—once at 4-bit and once at fp16—after observing unusual positional bias and below-published accuracy in the 4-bit run. The fp16 retry brought test-set accuracy from 0.536 to 0.571 and reduced positional skew from $2.3\times$ to $1.5\times$. The entropy coefficient moved from -0.010 ($p = 0.018$) to -0.004 (n.s.), and FDR rejection at $\alpha = 0.01$ moved from 46/57 to 55/57—closer to the larger models in our set. We interpret this as evidence that 4-bit quantization at the smallest scales amplifies positional priors and depresses confidence; the effect appears substantially weaker for 7B and 14B. We use the fp16 1.5B as the canonical 1.5B in the main analysis and treat the comparison as a precision-effect appendix.

Limitations. Several limitations warrant caution. First, domain-level variance is estimated at zero, likely because subject-level random effects absorb whatever between-domain structure exists; with only four domain groups, power to detect a separate domain component is very low, so we cannot cleanly partition between-subject variance into domain-level and subject-within-domain components. Second, the mixed model assumes homoskedastic residuals, which is violated when calibration gap variance depends on per-subject accuracy. Third, confidence is computed from logprobs at a single token position; this may not faithfully represent the model's internal uncertainty, especially after RLHF fine-tuning [Kadavath et al., 2022]. Fourth, results are specific to MMLU's four-choice format and may not generalize to open-ended generation settings. Fifth, the within-family scaling comparison is observational rather than controlled; the Qwen-0.5B endpoint is contaminated by extreme positional bias and does not contribute usable signal to the scaling comparison. Sixth, 4-bit quantization affects four of the five open-weight model runs and may modestly attenuate confidence; we conducted a precision-effect comparison at 1.5B but not at larger scales.

Future directions. Three directions extend the framework most directly. First, additional scale points within the Qwen family (e.g., Qwen-2.5-32B) would let us extend the monotonic ICC and entropy-coefficient progression beyond 14B. Second, adding a third model family (e.g., Mistral-7B or Gemma-2-9B at matched scale) would strengthen the cross-family generalization claim from two families to three. Third, applying the framework to MMLU-Pro or other harder benchmarks would test whether the high-school-vs-college calibration gap we observe within MMLU persists at higher difficulty levels.

7 Conclusion

We presented a statistical audit framework for LLM calibration heterogeneity and applied it to MMLU using six instruction-tuned models from two families across five parameter scales. The framework recovers a coherent contrast in where calibration succeeds versus fails: applied/factual high-school subjects are reliably well-calibrated, while college-level technical content, niche factual recall, and structured ethical/legal reasoning are reliably bad. Subject-level ECE rankings are highly correlated across the five larger model pairs (Spearman $\rho \in [0.78, 0.94]$); the smallest model in our set (Qwen-0.5B) shows extreme positional bias that produces a qualitatively different miscalibration pattern, which we interpret as a lower bound below which the elicitation protocol breaks down. Within the Qwen family at matched precision, both between-subject ICC and the entropy coefficient grow monotonically across the 1.5B/7B/14B scaling sweep—an observational pattern consistent with RLHF-induced confidence saturation rather than a controlled mechanism test. The audit's stable targets provide the structural justification for choosing subject-aware over global recalibration scoping—a choice operationalized in concurrent work by Luo et al. [2025]—and supply a transferable diagnostic for flagging miscalibration on new models before deployment.

Statement of Work

Data analysis. Jennifer Zhang conducted the original two-model analysis (GPT-4o-mini, Llama-3-8B-Instruct), including the multilevel model, calibration curves, FDR correction, and sensitivity checks. John Wright collected and analyzed the four additional Qwen-2.5 model runs (0.5B, 1.5B 4-bit, 1.5B fp16, 7B, 14B), extended the multilevel and cross-model framework to six models, and produced the within-family scale, entropy-mechanism, and quantization-effect analyses. **Coding.** Jennifer Zhang built the original pipeline; John Wright extended it to support arbitrary models and produced the parameterized inference notebook used for the open-weight models. **Writing.** Jennifer Zhang drafted the original report; John Wright revised it for the six-model expansion and produced the final manuscript. **Presentation.** Jennifer Zhang prepared the slide deck.

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